

TASK 4: EV DESIGN & ENERGY PROJECT

Final conceptual design for a compact urban electric passenger vehicle

Submitted by	Saurabh Saroj
Discipline	B.Tech – Electrical Engineering
Internship	Electric Vehicle Technology
Application	Compact 5-seat urban EV
Design focus	Battery • Motor • Range • Energy consumption

Scope: This is an internship-level conceptual design. Production EV development would require detailed vehicle dynamics, thermal, mechanical, safety and homologation validation.

1. Basic EV System Design

A compact 5-seat urban EV is selected for daily commuting, city traffic and occasional highway use. The system uses a high-voltage lithium-ion battery, inverter, traction motor, BMS, DC-DC converter and regenerative braking.

Subsystem	Selected design	Function
Battery pack	96S2P Li-ion, 355.2 V, 100 Ah	Stores traction energy
BMS / protection	BMS + contactor + fuse	Monitoring, balancing and protection
Inverter	High-voltage 3-phase inverter	Converts battery DC to controlled motor AC
Traction motor	50 kW peak / 25 kW continuous	Vehicle propulsion
DC-DC converter	HV DC → 12 V system	Auxiliary electrical loads
Regenerative braking	Motor operated as generator	Recovers part of braking energy

Energy flow: Battery → Inverter → Traction Motor → Wheels, with regenerative braking returning electrical energy toward the battery.

2. Battery Capacity Estimation

The battery is sized from the target operating range and energy consumption, then checked against the selected 96S2P pack.

Target range assumption: approximately 240 km at a baseline consumption of 130 Wh/km.

Required usable energy = $240 \times 0.130 = 31.20 \text{ kWh}$

With 90% usable fraction, required nominal energy = $31.20 \div 0.90 = 34.67 \text{ kWh}$

Selected pack energy = $355.2 \text{ V} \times 100 \text{ Ah} \div 1000 = 35.52 \text{ kWh}$

Usable selected energy = $35.52 \times 0.90 = 31.97 \text{ kWh}$

Therefore, the selected 35.52 kWh pack meets the conceptual 240 km target under the baseline assumption.

3. Motor Requirement Calculation

For a numerical engineering check, assume vehicle mass = 1500 kg, aerodynamic drag coefficient $C_d = 0.29$, frontal area = 2.2 m^2 , rolling coefficient $C_{rr} = 0.012$, air density = 1.225 kg/m^3 and drivetrain efficiency = 90%.

Check	Result	Design implication
80 km/h flat-road cruise	$\approx 9.1 \text{ kW}$ required	25 kW continuous rating is adequate
100 km/h flat-road cruise	$\approx 14.8 \text{ kW}$ required	Comfortable continuous margin
0–50 km/h in 10 s	$\approx 36.0 \text{ kW}$ at 50 km/h	50 kW peak provides acceleration margin
100 km/h on 5% grade	$\approx 37.5 \text{ kW}$	50 kW peak supports short gradients

Key equations: $F_{\text{total}} = F_{\text{rolling}} + F_{\text{aero}} + F_{\text{grade}} + F_{\text{acceleration}}$; $P = F \times v \div \eta$. The resulting 50 kW peak / 25 kW continuous motor is therefore technically justified for this conceptual vehicle.

4. Approximate Range & Energy Consumption

Range is estimated from usable battery energy divided by energy consumption. Consumption changes with speed, acceleration, road gradient, payload, HVAC use, tyre condition and regenerative braking.

Usage condition	Assumed consumption	Estimated range	Reason
Efficient city / light traffic	120 Wh/km	≈ 266 km	Lower speed + regenerative braking
Baseline urban / mixed	130 Wh/km	≈ 246 km	Primary design condition
Mixed / normal driving	140 Wh/km	≈ 228 km	Moderate traffic and varied speed
Highway / higher speed	170 Wh/km	≈ 188 km	Higher aerodynamic drag
Heavy load / AC / difficult	190 Wh/km	≈ 168 km	Higher traction + auxiliary demand

Baseline calculation: Range = 31.97 kWh ÷ 0.130 kWh/km ≈ **246 km**.

5. Energy Analysis Under Different Conditions

Condition	Main energy effect	Expected impact
City traffic	Frequent acceleration/deceleration	Moderate consumption; regenerative braking can recover part of braking energy
Highway	Aerodynamic drag rises strongly with speed	Higher Wh/km and reduced range
Uphill / hilly route	Potential-energy increase	High temporary power demand
Heavy payload	Higher rolling and acceleration force	Higher Wh/km
AC / heating	Additional auxiliary electrical load	Reduced driving range
Eco / smooth driving	Lower speed and gentler acceleration	Lower Wh/km and improved range

6. Final Design & Technical Justification

The final concept combines a 35.52 kWh nominal battery, 31.97 kWh usable energy, 50 kW peak / 25 kW continuous traction motor and an estimated baseline range of about 246 km. The battery is large enough to satisfy the calculated energy requirement for a ~240 km target while remaining moderate in size. The motor calculation shows that 25 kW is adequate for sustained cruising and 50 kW provides useful acceleration and grade capability.

Final conclusion: Battery capacity, motor power and vehicle energy consumption must be designed together. A larger battery increases range but also mass, cost and charging demand; a higher-power motor improves performance but can increase peak electrical demand. The proposed design is therefore a balanced compact-EV concept rather than a maximum-performance specification.

7. Final Design Summary

Parameter	Final value
Vehicle application	Compact 5-seat urban EV
Battery configuration	96S2P Li-ion
Nominal voltage	355.2 V
Battery capacity	100 Ah
Nominal energy	35.52 kWh
Usable energy assumption	31.97 kWh (90%)
Peak motor power	50 kW
Continuous motor power	25 kW
Baseline consumption	130 Wh/km
Baseline estimated range	≈ 246 km
Estimated range band	≈ 168–266 km depending on conditions

References & Notes

- U.S. Department of Energy / Alternative Fuels Data Center — EV and charging technology guidance.
- Basic vehicle-dynamics relationships for rolling resistance, aerodynamic drag, grade force and traction power.
- Battery and powertrain values are conceptual assumptions for internship analysis and are not a production-vehicle specification.
- The calculations are intended to demonstrate the design method requested in Task 4.